

SUMMARY

Ph.D. student in Computer Science (AI and Robotics) focused on curriculum learning and reinforcement learning for robot manipulation and navigation.

Developed GACL and Reward Training Wheels; 3 papers accepted at IROS 2025 and 1 paper in IEEE RA-L 2025.

Engineering experience at AWS building production-grade testing and monitoring systems that cut regression analysis time from 8 hours to 15 minutes.

EDUCATION

George Mason University <i>PhD in Computer Science, Specialization in AI and Robotics</i> - Research Focus: Curriculum Learning and Reinforcement Learning for Robotics - Relevant Coursework: Advanced Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision	Fairfax, VA <i>Sep 2023 – Present</i>
Carnegie Mellon University <i>MSc in Mechanical Engineering</i> - GPA: 3.94/4.0 (98.5%) - Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Deep Reinforcement Learning & Control	Pittsburgh, PA <i>Sep 2021 – May 2023</i>
University of Cincinnati <i>BS in Mechanical Engineering</i>	Cincinnati, OH <i>Sep 2016 – May 2021</i>

RESEARCH EXPERIENCE

RobotiXX Lab, George Mason University <i>Graduate Research Assistant, Curriculum Learning for Robotics</i> - Developed Grounded Adaptive Curriculum Learning (GACL) to generate task curricula grounded in robot performance feedback. - Implemented PPO and SAC in IsaacGym, achieving 24.58% higher success rate and 50% improved sample efficiency. - Designed Reward Training Wheels to adapt auxiliary rewards through learning for robust policy training. - Co-authored 3 IROS 2025 accepted papers and 1 IEEE RA-L 2025 publication.	Fairfax, VA <i>Aug 2023 – Present</i>
Computational Engineering and Robotics Lab, CMU <i>Research Assistant: 3D AR Scene Inpainting via Deep Learning</i> - Developed an end-to-end deep learning pipeline for 3D AR scene inpainting, achieving 92% accuracy in scene completion. - Implemented and fine-tuned a GAN model for image inpainting, improving texture realism by 35% over baseline methods. - Applied RANSAC and DBSCAN algorithms for efficient 3D point cloud segmentation, reducing processing time by 40%.	Pittsburgh, PA <i>Jan 2022 – May 2023</i>

INDUSTRY EXPERIENCE

Amazon Web Services (AWS) <i>Software Development Engineer Intern - RDS Proxy Team</i> - Built IPEBench Streamlit visualization suite (8 interactive views) cutting regression analysis time from 8 hours to 15 minutes. - Developed regression testing framework (10,000+ lines) with Welch’s t-test, power analysis, and Bonferroni correction for 99% confidence detection. - Implemented adaptive sampling with Thompson Sampling and Bayesian optimization, improving test reliability from 47% to 90% and reducing false positives. - Integrated CloudWatch metrics into automated dashboards for multi-region RDS Proxy performance monitoring.	Bellevue, WA <i>May 2025 – Aug 2025</i>
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TECHNICAL SKILLS

Programming Languages:	Python (Advanced), C++ (Intermediate), CUDA, SQL, Bash
Robotics & Simulation:	ROS, IsaacGym, MuJoCo, OpenAI Gym
ML/RL:	PyTorch, TensorFlow, JAX, PPO, SAC, Curriculum Learning
Systems & Cloud:	AWS (RDS, CloudWatch, EC2, Lambda, S3), Docker, Kubernetes
Tools & Libraries:	NumPy, Pandas, Matplotlib, OpenCV, Git, CI/CD
Statistical Analysis:	Hypothesis Testing, Power Analysis, A/B Testing, Regression Analysis

PUBLICATIONS

Wang, L., et al. "GACL: Grounded Adaptive Curriculum Learning"	IROS 2025 (Accepted)
Wang, L., et al. "Reward Training Wheels: Adaptive Auxiliary Rewards"	IROS 2025 (Accepted)
Lu, Y., et al. "Decremental Dynamics Planning for Robot Navigation"	IROS 2025 (Accepted)
Zhao, C., et al. "II-NVM: Normal Vector-Assisted Mapping"	IEEE RA-L 2025